universidade de aveiro
deti departamento de eletrónica, telecomunicações e informática

Machine Learning Operations

Software Engineering

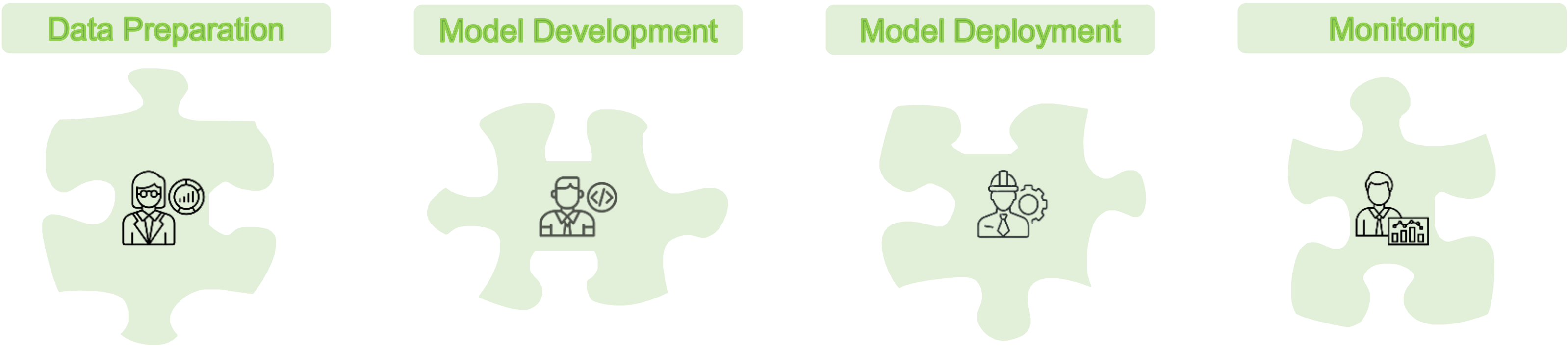
2025/2026

Outline

- What is MLOps?
- Lifecycle
- Integration
- LLMOps

Context

- Your system stores the conversations of users with the chatbot;
- These data is used by data scientists to retrain the chatbot model;
- How do we manage this in a production environment?



Code vs Models

The “gap”

DevOps

- **Versioning**: Code
- **CI/CD**: Software test and deploy
- **Monitoring**: Infrastucture and services

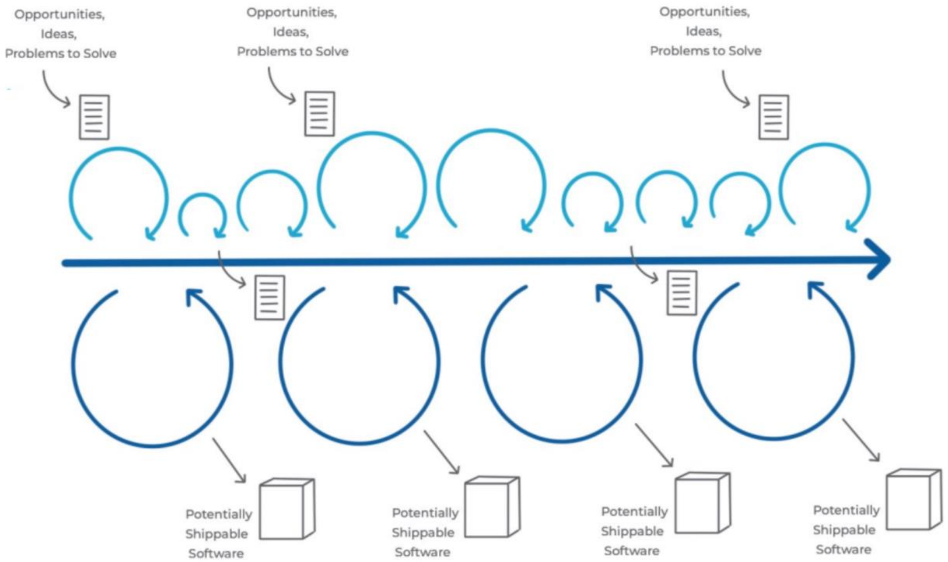
Model requirements

- **Versioning**: Code + **data** + **model**
- **CI/CD**: + **Continuous training (CT)**
- **Monitoring**: Infra/services + **model quality**

Continuous training

- The model learned from historical data;
  - But, reality changes, new terms and problems appear, the product is updated;

- If the model is not retrained with new data its eficacy will degrade over time;
  - So, there is a need to automate the training pipeline.



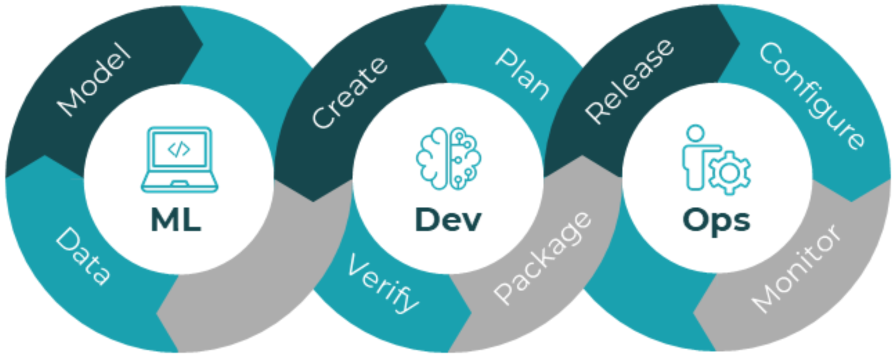
Quality monitoring

- The model is available;
- However, its previsions start having an high error rate;
- This is not detected by DevOps, it is an error in the model’s logic, not on its code or server.

Machine learning operations

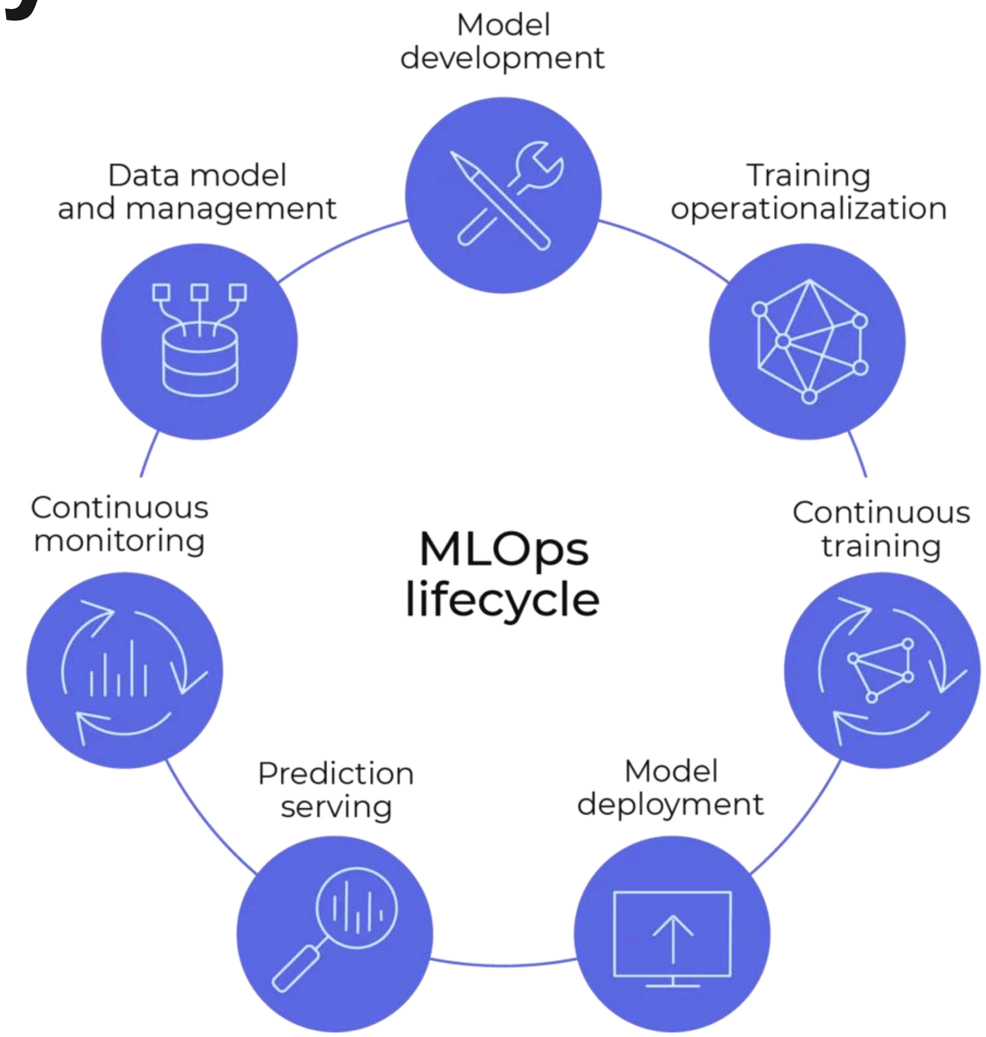
The solution

- MLOps is a set of practices that extends de DevOps principles to machine learning specific needs;
- Standardise and automate the entire ML lifecycle, from experimentation to production;
- Ensure the reliability, scalability, and maintainability of ML models in production environments.



Lifecycle

MLOps lifecycle
Stages



MLOps lifecycle

Data management

The foundation of any successful model lies in the quality and integrity of its data.

- Establishing pipelines, for data collection and processing;
- Implementing data versioning and lineage tracking;
- Data Version Control (DVC) ensures consistent, well-documented data.

MLOps lifecycle
Model development

Creation

Includes Exploratory Data Analysis (EDA), feature engineering, model training, and hyperparameter tuning.

Collaboration

Data scientists and MLOps engineers work together to ensure models meet performance criteria.

Management

Tools like MLflow and TensorFlow Extended (TFX) track experiments and package models.

MLOps lifecycle

Model deployment

- Moving models from development to production involves packaging the model and infrastructure setup (e.g., containers, Kubernetes clusters), and integrating the model with existing applications or services.

- Cloud services like AWS SageMaker, Google Cloud AI, and Azure ML simplify this process by providing managed services and streamlined deployment workflows

Amazon SageMaker

vertex.ai

Azure Machine Learning

MLOps lifecycle

Monitoring and maintenance

Deployment is not the end. We must track behavior over time to prevent degradation.

- **KPIs**: Monitor prediction accuracy, response times, and potential bias.
- **Drift detection**: Identifying changes in data patterns that affect performance.
- **Maintenance**: Periodic retraining and fine-tuning to keep models aligned with reality.

MLOps lifecycle

Retiring and replacing

Over time, models may become outdated or less effective due to changes in data patterns, business requirements, or technological advancements.

- **Assess**

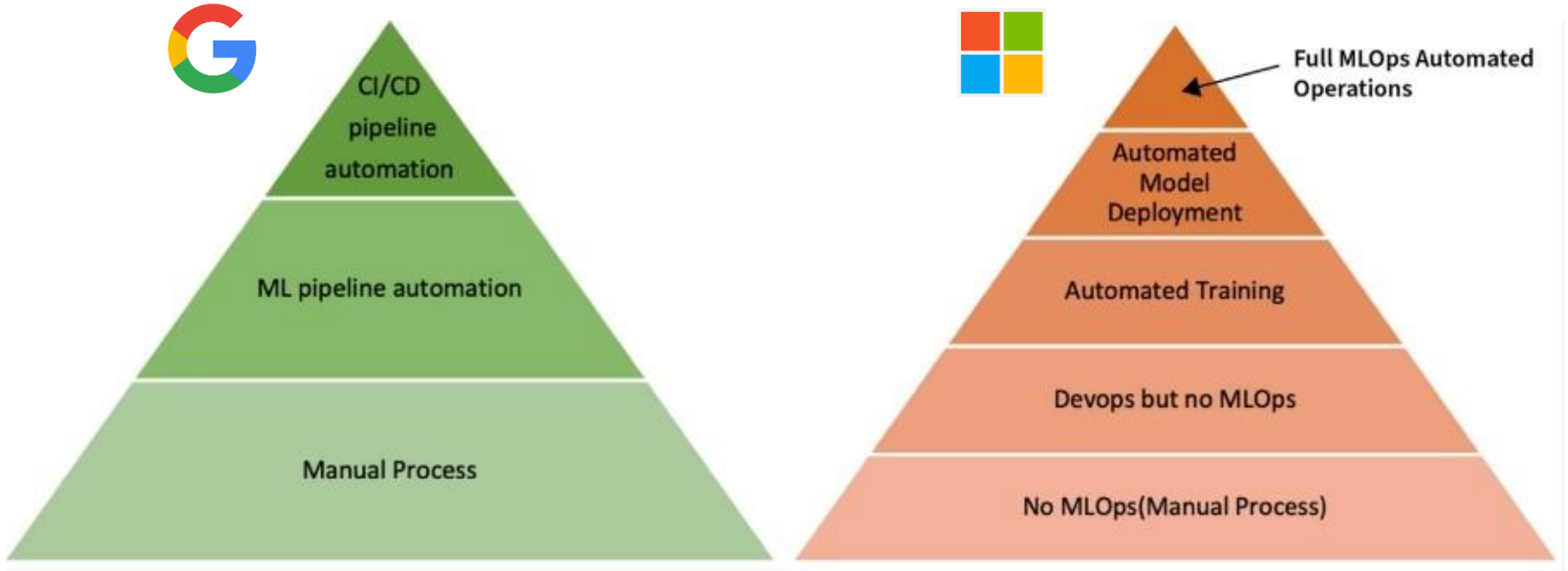
  Determine if the model is outdated due to new data patterns or tech advancements.

- **Replace**

  Execute the transition to the improved model version, while minimizing production disruption.

Integration

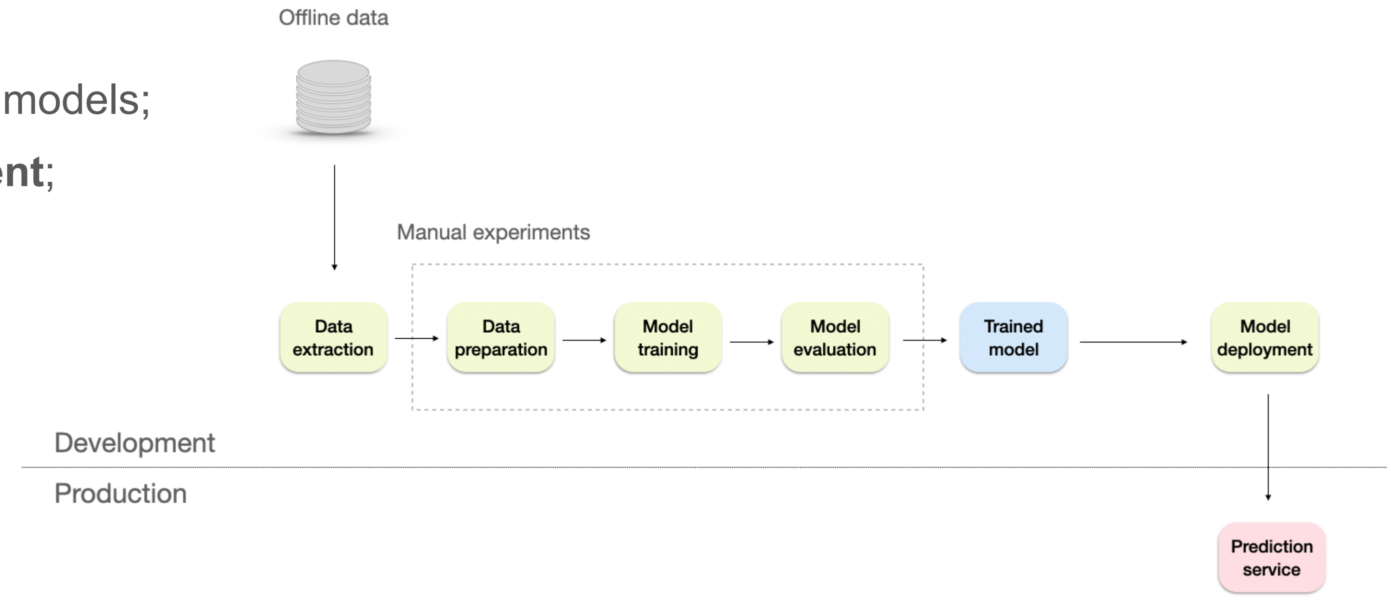
MLOps implementation
Maturity levels



MLOps maturity levels

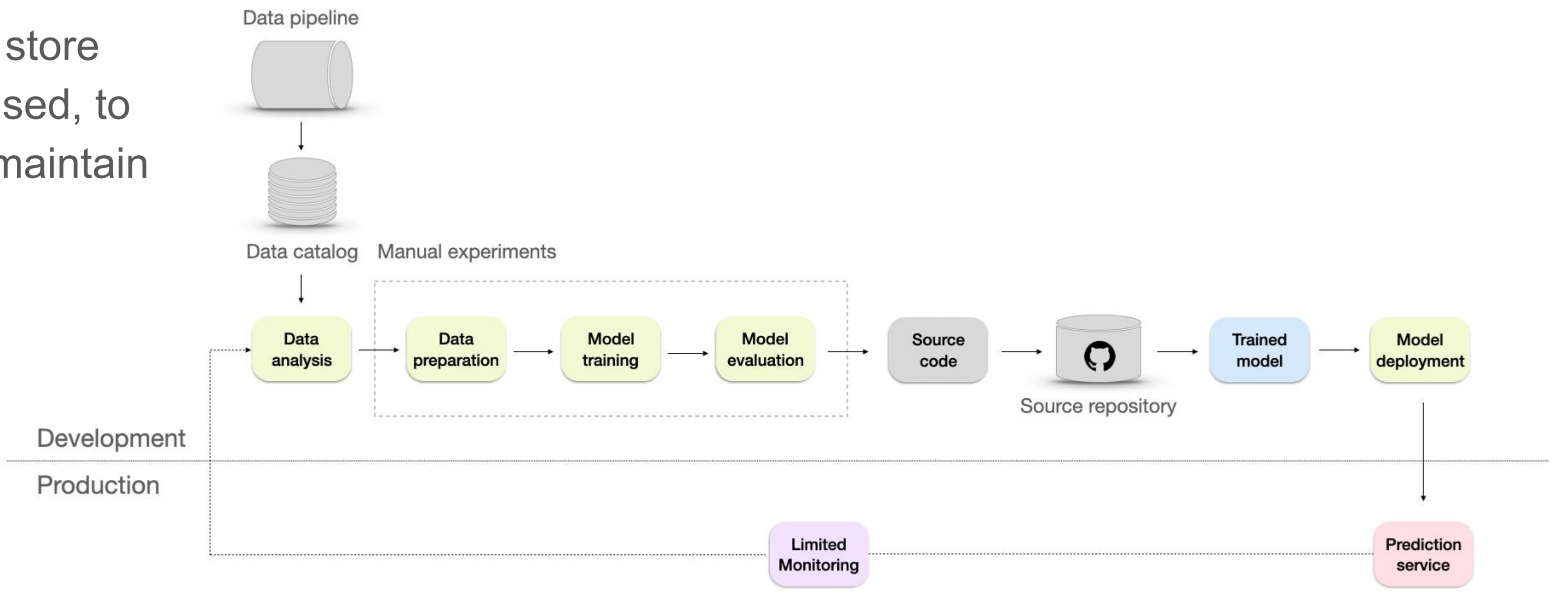
Level 1

- Data processing, experimentation, and model deployment processes are **entirely manual**;
- Workflow relies heavily on data scientists, with some assistance from a data engineer to prepare the data and a software engineer to integrate the model with the product/business processes;
- Applied in early-stage starts-ups and proof of concept projects, small-scale ML applications, and ad hoc ML tasks;
- **No frequent retrains** of production models;
- Releases are **difficult** and **infrequent**;
- **No versioning**.



MLOps maturity levels
Level 2

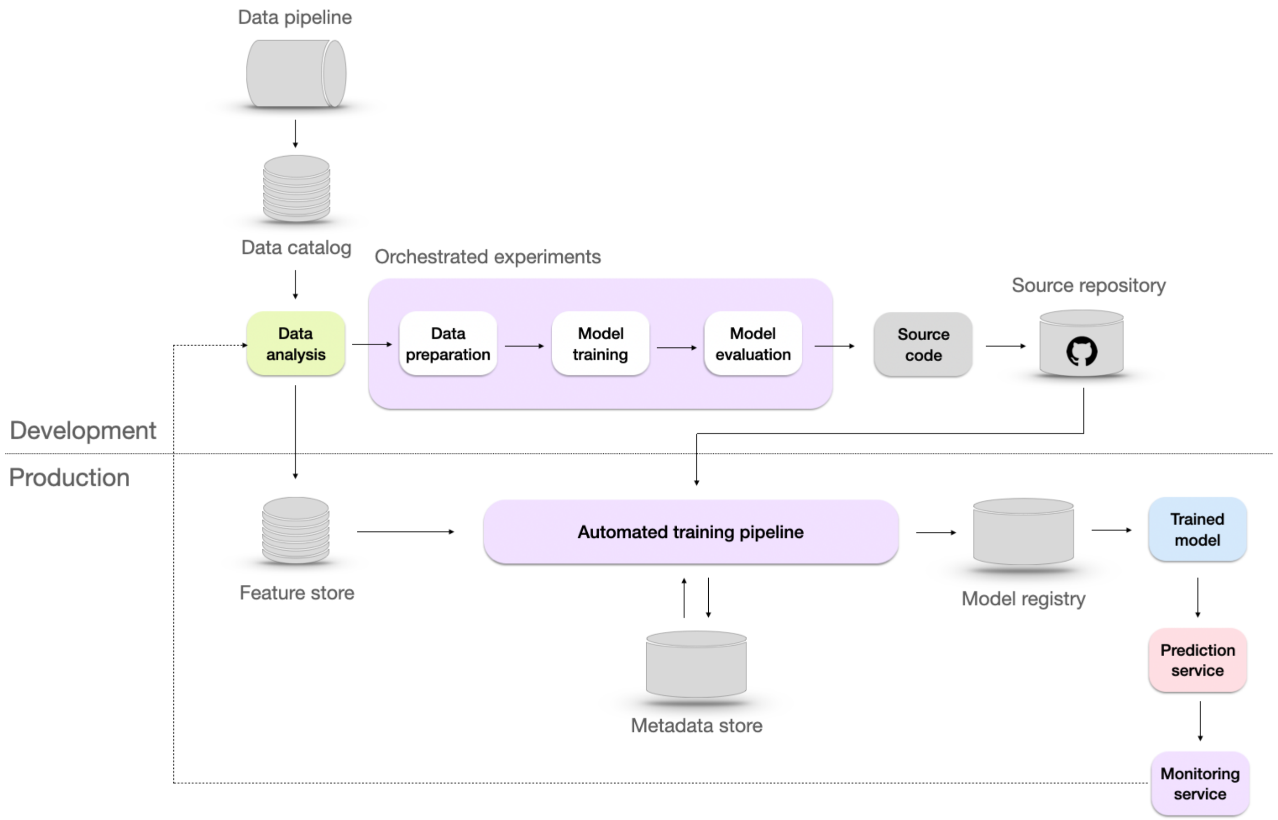
- Introduction of **DevOps practices** to the infrastructure by converting the experiments to the source code and storing them in the source repository;
- **Automated data pipeline** to extract the data from different sources and combine them together, improving scalability, efficienty, and accuracy;
- Addition of a **data catalog** to store metadata from the datasets used, to help organize, manage, and maintain the large volumes of data.



MLOps maturity levels

Level 3

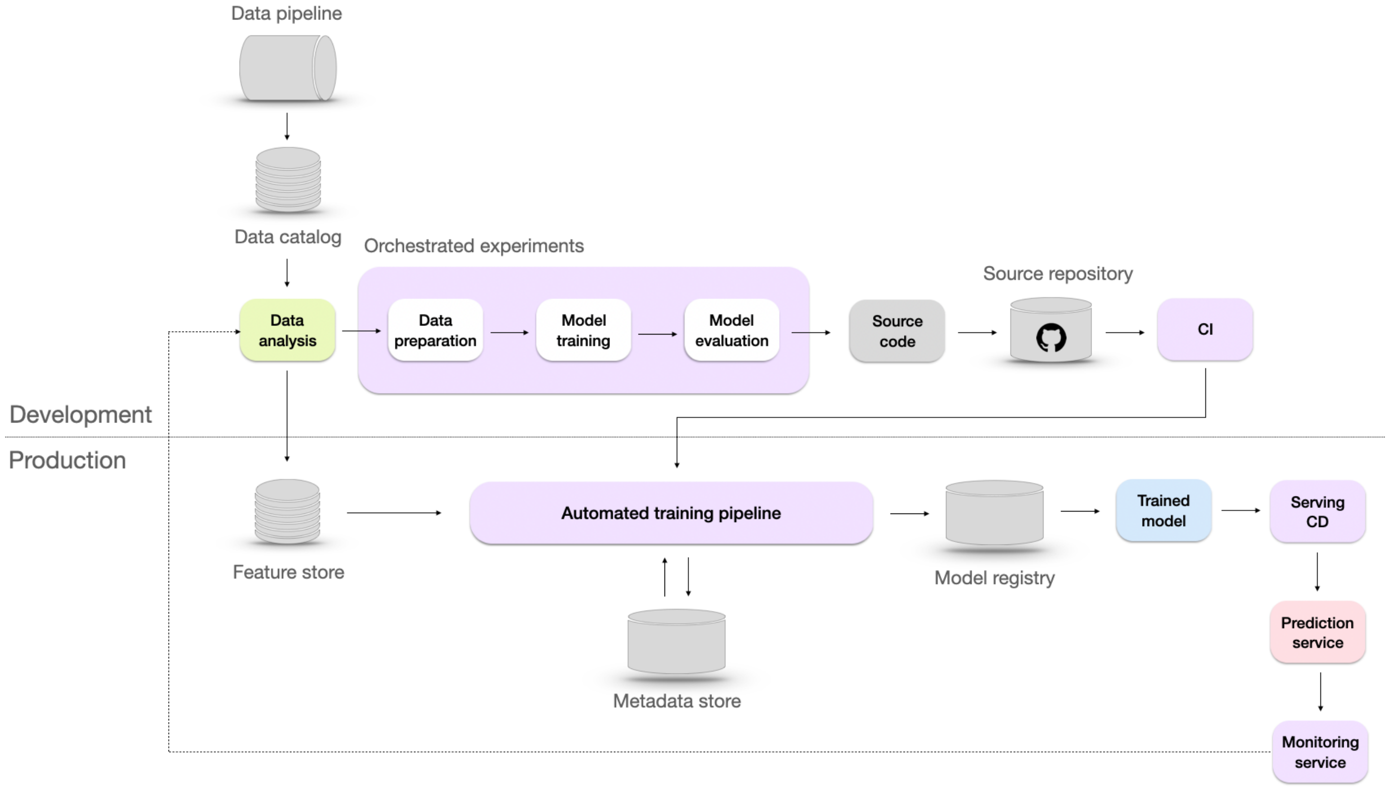
- **Automated training pipeline** to handle the end-to-end process of training models, from data preparation to model evaluation;
- A **metadata store** to track and manage information of all the experiments data;
- A **model registry**, which is a repository to store ML models, their versions, and their artifacts;
- A **feature store** to provide a centralized location for feature storing, managing, serving, and evolution tracking over time.



MLOps maturity levels

Level 4

- Automation of the building, testing, and deployment stages;
- Integration of **CI/CD pipelines** to allow for rapid deployment of model updates, improvements, and bug fixes;
- Model validation with **A/B testing**, which involves comparing predictions and user feedback between an existing model and a candidate model to determine the better one.

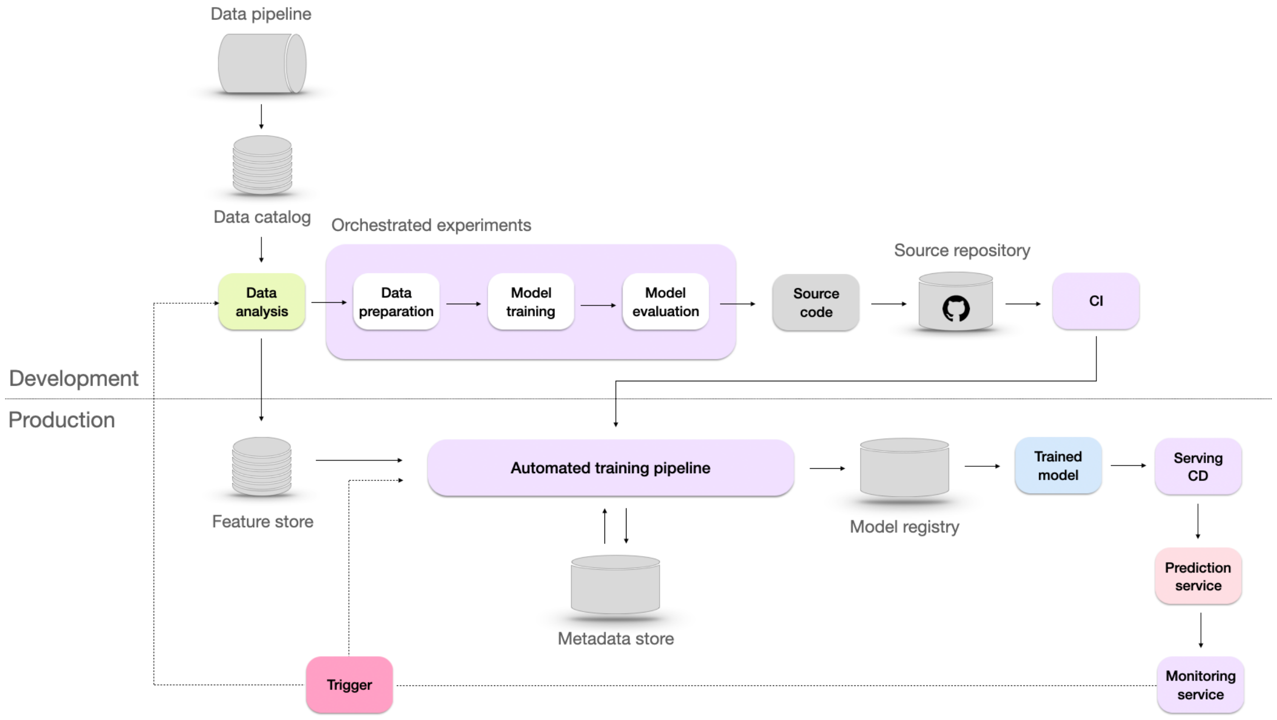


MLOps maturity levels

Level 5

- The model is **automatically retrained** based on the trigger from the monitoring system;
- Ensure the model **remains effective** even when faced with unexpected changes in the data (data drift)

- **Adapt to rare events** such as Black Friday, where patterns and trends in the data may significantly deviate from the norm;
- **Minimize the cold start problem**, which arises when the model needs to make predictions for new users lacking historical data



MLOps maturity levels

Considerations

- Higher maturity levels don't automatically translate to better business results

- Don't follow maturity level guidelines blindly since vendors' main incentive is selling services
  - For example, continuous retraining requires significant computation power, but is often unncessary or even harmful; it should be triggered by actual need.

- Focus on reliable monitoring systems and effective root cause analysis processes

Large Language Model Operations

MLOps vs LLMOps

The “gap”

MLOps

- **Main asset:** Trained model
- **Training**: Epochs and versioned data
- **Monitoring**: Prediction accuracy

LLM requirements

- **Main asset:** Model + **prompts** + **context**
- **Training**: *Fine-tuning* or *In-context learning*
- **Monitoring**: Generation quality

Key stages of the LLMOps lifecycle

Prompt and context engineering

➢ Prompt engineering

Developing and testing the ideal prompts for the specific use case. These prompts must be versioned just like code.

➢ Context management (RAG - *Retrieval Augmented Generation*)

Instead of training the LLM on internal company data (as would be the case in classic MLOps), relevant information is provided from a vector database.

Challenge: It is necessary to build and maintain the ingestion and update pipeline for the vector database, ensuring its traceability and consistency.

Key stages of the LLMOps lifecycle

Deployment and inference optimization

- **Service**

Ensuring that the model (whether hosted or accessed via a third-party API) is invoked resiliently, and that latency (time to first token and total time) is minimized.

- **A/B testing**

Essential for testing different versions of prompts or models before exposing them to all users.

Key stages of the LLMOps lifecycle

Monitoring and security

➢ Quality monitoring

Evaluating the output using metrics that measure toxicity, tone, relevance, and hallucinations.

This often requires a secondary model (model-as-a-judge) or human evaluators.

➢ Security guardrails

Implementing filtering layers (input and output) to prevent attacks or policy violations.

➢ Data drift in LLMs

Monitoring changes in the distribution of user prompts over time, which may indicate the need to adjust internal prompts or the RAG system.

Bibliography